

Reputation Model

Overview =>

The reputation model defines how a user's trust score is calculated, updated, and maintained over time. Reputation represents a numerical estimate of a user's reliability based on interaction history and inactivity. The model is deterministic (**for now**) meaning the same inputs always produce the same outputs.

(for now→ later we will be looking for a non deterministic model as in for consequences based loss)

Reputation Range =>

Reputation values are bounded within a fixed range from 0(zero) to 100, no negative(**for now**) to prevent uncontrolled growth or negative values.

(for now→ later we might decide for a negative score too for longer stay on zero, longer stagnant on zero will take more time/rule following to get back to positive)

Initial Reputation =>

When a user interacts with the system for the first time, an initial reputation value is assigned(neutral starting value)(**for now**). This ensures new users are neither fully trusted nor fully distrusted.

(for now→ later we start them as untrusted, they need to maintain high activity to get to neutral)

Interaction-Based Increase =>

When a user performs a valid interaction, reputation increases by a fixed reward value for that interaction. Reputation must not exceed the maximum limit. This rewards active participation.

Time-Based Decay =>

Reputation decreases if a user remains inactive. This ensures reputation reflects recent activity. Key rules:

- Decay is based on time elapsed since the last interaction.
- Decay occurs in fixed intervals.
- Example: -5 points per 30 days of inactivity.
- Reputation cannot go below the minimum limit.

Decay Application Rule => (flaw)

Decay is applied only when a user interacts again. This design choice exists because smart contracts cannot execute automatically without a transaction. Blockchain systems require a trigger to update state. When a user interacts:

1. The system calculates time since last update.
2. Decay is applied for elapsed intervals.
3. Interaction reward is then added.
4. The last update timestamp is refreshed.

Deterministic Behavior =>

For a guaranteed predictable and auditable behavior. This model does not:

- Use randomness.
- Use machine learning.
- Depend on external data sources.

All calculations depend only on:

- Stored reputation value.
- Stored last interaction timestamp.
- Current block timestamp.

Example Scenario =>

Initial reputation: 50

The user interacts after 60 days of inactivity.

Decay rule: -5 per 30 days.

60 days → 2 intervals → -10 decay.

Step-by-step update:

1. Apply -10 decay → 40
2. Apply +5 interaction reward → 45
3. Update timestamp

Final reputation: 45

